# ALEKSANDR HOVSEPYAN

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## PROFILE

Digital Forensics and Cyber Security graduate with hands-on experience across application security, digital forensics, AWS cloud security and machine learning for threat detection. Active CTF competitor seeking a graduate role in SOC operations, security engineering or threat detection.

## EDUCATION

**BSc (Hons) in Digital Forensics and Cyber Security** 2022-2026
*Technological University Dublin*
**Relevant Modules:** Secure Programming, Secure Communications, Computer and Network Forensics, Network Security, Professional Penetration Testing, Biometrics and Forensics Applications, Application Security, Network Security Analysis, Business Continuity Management and Cloud Security.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, SQL, Bash
**Security & Pen Testing:** Burp Suite, OWASP ZAP, Snyk, nmap, Wireshark, Metasploit, Kali Linux, T-Pot.
**Digital Forensics:** Autopsy, FTK Imager, iLEAPP, EXIF analysis, chain of custody
**Cloud & Infrastructure:** AWS (EC2, S3, IAM, VPC, CloudWatch, DLM, Elastic Beanstalk), Docker, Linux, Git
**Machine Learning:** XGBoost, Random Forest, scikit-learn, pandas, NumPy, feature engineering, OpenAI API / LLM integration
**Frameworks & Standards:** MITRE ATT&CK, NIST SP 800-86, NIST SP 800-101, ISO/IEC 27037, OWASP Top 10

## PROJECTS

**Phishing URL Detection — Chrome Extension (Final Year Individual Project)** 2025 – 2026
- Built a real-time phishing detection Chrome extension powered by XGBoost trained on the PhreshPhish dataset (666,000 URLs) using 36 lexical features selected via SHAP-based reduction from an initial 40.
- Achieved F1 = 0.946 on a 168,000-URL test set with 0.7 microsecond per-URL inference; tuned the decision threshold to a 1% false positive rate for browser deployment and benchmarked against Random Forest baseline on a Kaggle GPU pipeline.

**Cloud Resilience Architecture — AWS BCP / Disaster Recovery** 2026
- Designed and deployed a Pilot Light disaster recovery architecture across two AWS regions (eu-west-1 primary, eu-west-2 DR) meeting RTO 4hr / RPO 1hr.
- Configured EC2, S3 versioning with cross-region replication, IAM least-privilege roles, DLM snapshots, and AMI replication; validated failover with a Python chaos script.

**Digital Forensic Investigation — Windows XP Extortion Case** 2026
- Conducted full forensic analysis of three evidence images using Autopsy and FTK Imager; recovered deleted files and parsed Outlook Express mailboxes.
- Validated EXIF-tagged surveillance photos to establish premeditation; produced a chain of custody report aligned to NIST SP 800-86.

**ReconBot — OSINT Discord Bot (Team of 3)** 2025
- Co-developed a Python Discord bot delivering eight OSINT slash commands (breach detection, IP reputation, WHOIS, port scanning, Google Dorking) integrating HaveIBeenPwned, AbuseIPDB, LeakCheck APIs and OpenAI GPT-4o mini for AI-driven result explanation.
- Designed and built the password breach module and the IP reputation module with colour-coded threat scoring; owned the async cog-based system architecture and all functional, security, integration, and use-case testing.

**Distributed Honeypot Infrastructure — T-Pot** 2025
- Deployed a distributed T-Pot honeypot infrastructure across DigitalOcean and GCP in a hive-and-sensor configuration, capturing live attacker data on SSH, HTTP, SMB and industrial protocols.
- Monitored attack patterns, source IPs and tooling via the integrated ELK stack and Cowrie/Dionaea sensors; analysed scanning campaigns and brute-force activity against the exposed sensor nodes.

## EXPERIENCE

**Sales Assistant — Best Menswear** October 2024 – January 2025
- Delivered customer service in a fast paced retail environment; worked across rotating shifts with a small team during seasonal peaks, developing communication, reliability and time-management skills.

## CTF COMPETITIONS & ACTIVITIES

**ZeroDays CTF 2026 — Team Fantastic4-1 (Croke Park)** March 2026
- Placed 33rd of 147 teams overall and 15th in the college category

## CERTIFICATIONS

- Google Chronicle SOAR Fundamentals
- AttackIQ MITRE ATT&CK Fundamentals